



U.S. Department
of Transportation

**Pipeline and Hazardous
Materials Safety
Administration**

August 20, 2021

1200 New Jersey Avenue, SE
Washington, DC 20590

DOT-SP 8795
(FOURTEENTH REVISION)

EXPIRATION DATE: 2025-07-31

(FOR RENEWAL, SEE 49 CFR 107.109)

1. GRANTEE: Norris Cylinder Company
Longview, TX
2. PURPOSE AND LIMITATIONS:
 - a. This special permit authorizes the manufacture, mark, sale and use of non-DOT specification cylinder conforming with all regulations applicable to a DOT specification 4B240ET, except as specified herein, for the transportation in commerce of the hazardous materials authorized by this special permit. This special permit provides no relief from any Hazardous Materials Regulation (HMR) other than as specifically stated herein. The most recent revision supersedes all previous revisions.
 - b. The safety analyses performed in development of this special permit only considered the hazards and risks associated with transportation in commerce.
 - c. In accordance with § 107.107(a), party status may not be granted to a manufacturing special permit. The packaging authorized under the terms of this special permit may be used in accordance with § 173.22a.
3. REGULATORY SYSTEM AFFECTED: 49 CFR Parts 106, 107 and 171- 180.
4. REGULATIONS FROM WHICH EXEMPTED: 49 CFR §§ 173.304(a) and 173.309(a) in that non-DOT specification cylinders are not authorized, except as specified herein.
5. BASIS: This special permit is based on the application of Norris Cylinder Company dated August 6, 2021, submitted in accordance with § 107.109.

6. HAZARDOUS MATERIALS (49 CFR 172.101):

Hazardous Material Description			
Proper Shipping Name	Hazard Class/ Division	Identification Number	Packing Group
Bromotrifluoromethane or Refrigerant gas, R13B1	2.2	UN1009	N/A
Fire extinguishers containing compressed or liquefied gas	2.2	UN1044	N/A

7. SAFETY CONTROL MEASURES:

PACKAGING: Packaging prescribed is a non-DOT specification cylinder made in conformance with DOT Specification 4B240ET (§§ 178.35 and 178.55) except as follows:

§ 178.35(c) *Duties of inspector.*

(Add) (5) Prior to initial production of any design or design change, verify that the design qualification tests prescribed in § 178.55(d) of this special permit have been performed with acceptable results.

§ 178.35(f) *Marking.*

(1) Each cylinder must be plainly and permanently marked by stamping on the cylinder shoulder, top head, or neck as follows:

(i) DOT-SP 8795 followed by the service pressure and the letters “ET” for example DOT-SP 8795 -350ET.

(ii) Marking must comply with § 178.35(f) except as specified above.

(2) Marking stamped into the sidewalls of cylinders having a service pressure of 150 psi or less is permitted only if all of the following conditions are met:

(i) Wall stress at test pressure may not exceed 24,000 psi.

(ii) Minimum wall thickness may not be less than 0.090”.

- (iii) Depth of stamping may not be greater than 15 percent of the minimum wall thickness, but at no time may it exceed 0.015”.
- (iv) Maximum outside diameter of cylinder may not exceed 5”.
- (v) If the carbon content exceeds 0.25 percent, the cylinder must be stamped before the required heat treatment.
- (vi) Stamping must be adjacent to top head.

§ 178.55(a) *Type, spinning process, size and service pressure.* Cylinders must be made from electric resistance welded tubing complying with § 178.55(b) of this special permit. Cylinders closed by spinning process are authorized. Water capacity may not exceed 180 pounds (4,900 cubic inches) nominal; service pressure may not exceed 500 psi.

§ 178.55(b) *Steel.* Open hearth, basic oxygen, or electric steel of uniform quality in compliance with ASTM A53 or A135 specification or as specified in 49 CFR Part 178, Appendix A, Table I is authorized.

§ 178.55(d) *Manufacture.* Heads attached by welding or brazing are not authorized.

- (1) Circumferential seams are not allowed.
- (2) * * *
- (3) Tubing not in compliance with ASTM pressure application must be qualified in accordance with ASTM pressure tube specification A135.

(Add)

- (4) Each new design and any significant change to any acceptable design must be qualified for production by testing prototype samples as follows:
 - (i) Twelve cylinders must be fabricated to the proposed design. These cylinders must be subjected to the tests and inspections prescribed in this special permit without any failure.
 - (ii) Three of the above cylinders must be pressure cycled from less than 100 psig to the minimum prescribed test pressure (see § 178.55-14(b) of this special permit) at a rate not to exceed 10 cycles per minute for 10,000 cycles without any failure.

(iii) A copy of the design qualification test results must be submitted to the Office of Hazardous Materials Safety (OHMS) prior to the first shipment.

(5) A significant change means a 10% or greater change in cylinder wall thickness, service pressure or diameter; a 50 percent change in water capacity provided that the diameter, steel and wall thickness remains the same; or any change in material or end design.

(6) Material furnished by new suppliers must be subjected to vendor qualification testing sequence furnished to OHMS in the September 29, 1982 letter.

§ 178.55(f) *Wall thickness.*

(1) The wall stress may not exceed 35,000 pounds per square inch for cylinders made from Appendix A Table I ASTM A53 or A135 specification steels. Minimum wall 0.090" for any cylinder over 6" outside diameter.

(2) Calculation must be made by the formula:

$$S = [P(1.3D^2 + 0.4d^2)]/(D^2 - d^2), \text{ where}$$

S = wall stress in pounds per square inch

P = minimum test pressure prescribed for water jacket test or 450 pounds per square inch whichever is the greater

D = outside diameter in inches

d = inside diameter in inches

§ 178.55(g) *Heat treatment.*

(1) The completed cylinders must be uniformly and properly heat-treated prior to tests and after all forming and welding operations.

(i) Cylinders made from ASTM A53 or A135 steel with integral formed heads or bases must be subjected to a normalizing operation. Normalizing and brazing operations may be combined provided the operation is carried out at a temperature in excess of the upper critical temperature of the steel.

(ii) Cylinders made from other materials must be heat treated by any suitable method in excess of 1100 °F except that liquid quenching is not permitted.

(2) Heat treatment is not required after welding weldable low carbon steel parts to attachments of similar materials which have been previously welded to the cylinder and properly heat-treated, if such subsequent welding does not produce a temperature in excess of 400 °F on the cylinder.

§ 178.55(h) *Openings in cylinders.*

(1) Openings are permitted only in the heads and must be centered.

(2) Size of opening may not exceed one-half of the outside diameter, nor 2.625 inches, whichever is smaller.

(3) All openings must be threaded. Threads must be in compliance with the following:

(i) Each thread must be clean cut, even, without any checks, and to gage.

(ii) Taper threads, when used, must be the American Standard Pipe Thread (NPT) type in compliance with Federal Standard H-28/7; or the National Standard Gas Taper Thread (NGT) type in compliance with Federal Standard H-28/9.

(iii) Straight threads, when used, must be the National Gas Straight Thread (NGS) type in compliance with Federal Standard H-28/9.

§ 178.55(i) *Hydrostatic test.*

(1) through (4) * * *

(5) Each 500 cylinders or fewer successively produced each day must constitute a test lot for the purpose of the burst test. One cylinder selected from each lot must be pressurized to destruction. If test cylinder bursts below 4 times the service pressure, or if cylinder passes the burst test but failure initiates in the weld or heat affected zone thereof the test lot must be rejected.

(i) If failure initiates in a localized defect away from any weld, 2 additional samples may be taken from the same lot and must be subjected to the burst test. If either cylinder fails, the lot must be rejected.

(ii) The rejected lot may be requalified without reheat treatment by testing each cylinder in accordance with the nondestructive test in ASTM A135 paragraph 11.

8. SPECIAL PROVISIONS:

a. In accordance with the provisions of Paragraph (b) of § 173.22a, persons may use the packaging authorized by this special permit for the transportation of the hazardous materials specified in paragraph 6, only in conformance with the terms of this special permit.

b. A person who is not a holder of this special permit, but receives a package covered by this special permit, may reoffer it for transportation provided no modification or change is made to the package and it is offered for transportation in conformance with this special permit and the HMR.

c. A current copy of this special permit must be maintained at each facility where the package is offered or reoffered for transportation.

d. Each packaging manufactured under the authority of this special permit must be either (1) marked with the name of the manufacturer and location (city and state) of the facility at which it is manufactured or (2) marked with a registration symbol designated by the Office of Hazardous Materials Safety for a specific manufacturing facility.

e. A current copy of this special permit must be maintained at each facility where the package is manufactured under this special permit. It must be made available to a DOT representative upon request.

9. MODES OF TRANSPORTATION AUTHORIZED: Motor vehicle, rail freight, cargo vessel, cargo-only aircraft, and passenger-carrying aircraft.

10. MODAL REQUIREMENTS: A current copy of this special permit must be carried aboard each cargo vessel or aircraft used to transport packages covered by this special permit. The shipper must furnish a current copy of this special permit to the air carrier before or at the time the shipment is tendered.

11. COMPLIANCE: Failure by a person to comply with any of the following may result in suspension or revocation of this special permit and penalties prescribed by the Federal hazardous materials transportation law, 49 U.S.C. 5101 et seq:

- o All terms and conditions prescribed in this special permit and the Hazardous Materials Regulations, Parts 171-180.
- o Persons operating under the terms of this special permit must comply with the security plan requirement.
- o Registration required by § 107.601 et seq., when applicable.

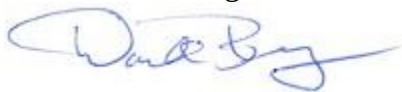
Each “Hazmat employee”, as defined in § 171.8, who performs a function subject to this special permit must receive training on the requirements and conditions of this special permit in addition to the training required by §§ 172.700 through 172.704.

No person may use or apply this special permit, including display of its number, when the special permit has expired or is otherwise no longer in effect.

Under Title VII of the Safe, Accountable, Flexible, Efficient Transportation Equity Act: A Legacy for Users (SAFETEA-LU)- “The Hazardous Materials Safety and Security Reauthorization Act of 2005” (Pub. L. 109-59), 119 Stat. 1144 (August 10, 2005), amended the Federal hazardous materials transportation law by changing the term “exemption” to “special permit” and authorizes a special permit to be granted up to two years for new special permits and up to four years for renewals.

12. REPORTING REQUIREMENTS: Shipments or operations conducted under this special permit are subject to the Hazardous Materials Incident Reporting requirements specified in 49 CFR §§ 171.15 Immediate notice of certain hazardous materials incidents, and 171.16 Detailed hazardous materials incident reports. In addition, the grantee(s) of this special permit must notify the Associate Administrator for Hazardous Materials Safety, in writing, of any incident involving a package, shipment or operation conducted under terms of this special permit.

Issued in Washington, D.C.:



for William Schoonover
Associate Administrator for Hazardous Materials Safety

Address all inquiries to: Associate Administrator for Hazardous Materials Safety, Pipeline and Hazardous Materials Safety Administration, Department of Transportation, Washington, D.C. 20590. Attention: PHH-13.

Copies of this special permit may be obtained by accessing the Hazardous Materials Safety Homepage at <https://www.phmsa.dot.gov/approvals-and-permits/hazmat/special-permits-search>. Photo reproductions and legible reductions of this special permit are permitted. Any alteration of this special permit is prohibited.

PO: KFW/NICKS